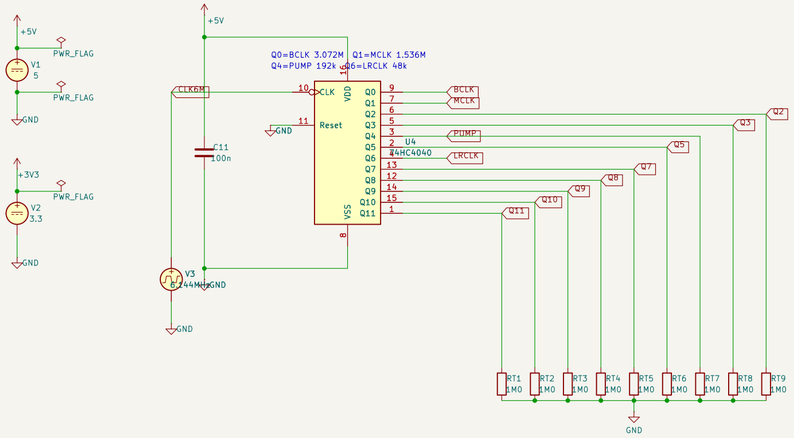
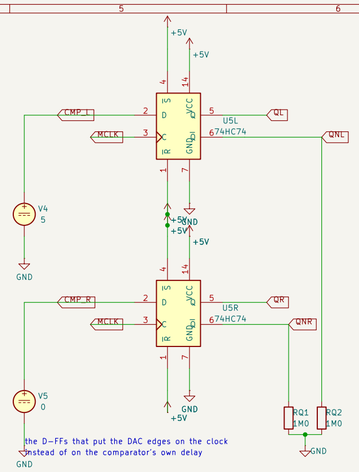
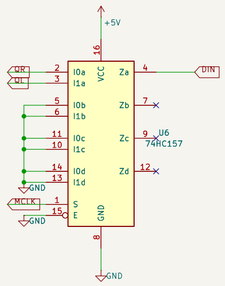
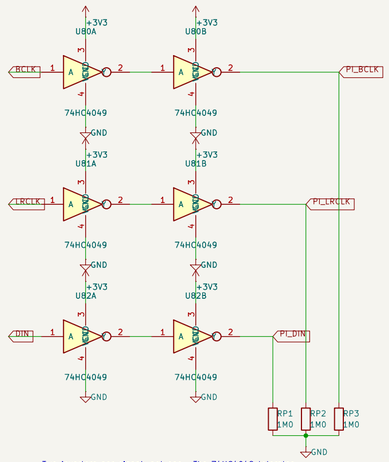
G The Pi interface (retime −> interleave −> 3.3 V: is there any setup margin left?)





MCLK selects: DIN carries R,L,R,L... at 3.072 Mbps.
64 BCLK per LRCLK frame = 32 L bits + 32 R bits
= exactly one OSR=32 sample per channel.





Two inverters per signal, not one. The 74HC4049 tolerates
an input above its own VCC −− it has no clamp diode to VCC −−
which is what makes the 5 V −> 3.3 V shift legal, but
inverting BCLK would make the Pi sample on the data
transition. 1M stands in for a GPIO input.

The claim under test: DIN changes on a BCLK falling edge and
the Pi samples it on the rising one, so half a BCLK period −−
163 ns −− is the setup margin, with no retiming flip-flop
needed anywhere. What eats into it is skew: MCLK is one
ripple stage further down the counter than BCLK, then the mux
and two 4049 stages add their own, while PI_BCLK goes through
two 4049 stages of its own. Only the DIFFERENCE matters, and
that is what this sheet measures.
L is held high and R low, so DIN must come out as a clean
1.536 MHz square; any other pattern means the two channels
are not landing in alternate bit slots.

```
.tran 2n 100u uic
```

<− back to the board schematic vinyl_adc.kicad_sch

Sheet: /
File: sim_g_interface.kicad_sch

Title: G − retiming, interleave and the 3.3 V level shift

Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 1/1